Advanced Optical Fiber Communications

Problem 13: Butt coupling of fibers

You want to build a pump-combiner for an Erbium-doped fiber amplifier. Due to the lower noise figure you decide to pump the Amplifier @ 980 nm.

a) You notice, your Pump Laser uses an Hi1060 fiber, while your Pump combiner uses an SMF-28e+ fiber. How big will the coupling loss of the splice be, when your fiber splicer can guarantee a lateral offset of 0.5 µm. fundamental mode only.
Give the coupling loss in dB.

b) Only the core of the gain fiber is Rare-earth doped. Check how much of the Field will interact with the Erbium Ions, if the Gain Fiber is equal in Diameter to the SMF28e+.

c) When a fused splice isn't possible and you need a mechanical connection (e.g., fiber connectors with an air gap), calculate the Fresnel reflection loss at the fiber-air interfaces at 980 nm. Assume normal incidence and a refractive index of n = 1.45 for silica at 980 nm. How much total insertion loss does this add compared to a fused splice?
Which other effects can occur with an existing airgap?

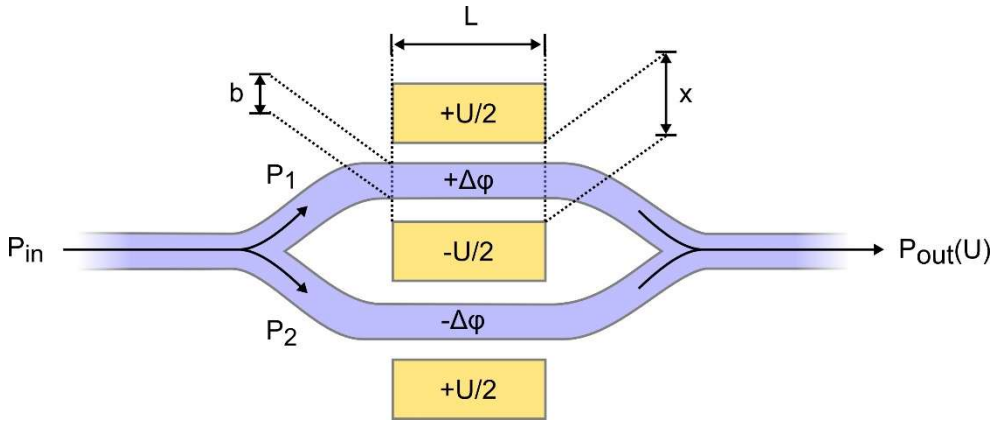
Now all modes inside the fiber will be considered

d) The SMF-28e+ fiber is designed for single-mode operation at 1310 nm and 1550 nm, with a specified cutoff wavelength of ≤1260 nm. Using the fiber parameters of core radius a = 4.1 µm and numerical aperture NA = 0.12, calculate the V-number at the 980 nm pump wavelength and determine which LP modes can propagate. Estimate the total number of guided modes and discuss the implications for using this fiber in an EDFA pump path.

e)

Problem 14: Mach-Zehnder modulator

The following figure represents the structure of a $LiNbO_3$ Mach-Zehnder modulator. The incident light power P_{in} is divided into two planar waveguides, which are controlled by phase modulators. P_1 and P_2 correspond to the power of light and $\Delta\varphi$ and $-\Delta\varphi$ are the introduced phase shifts in each branch, respectively.



a) Express the transmission T at the output of the Mach-Zehnder modulator as a function of the ratio $p = P_1/P_2$ and the phase shift $\Delta\varphi$. Assume a zero-phase difference for the case where no external voltage is applied. Deduce then the maximum and minimum transmission T_{max} and T_{min} as a function of p.

The following diagrams show the measured optical output power of the Mach-Zehnder modulator for an input power of 200 µW in a linear and a logarithmic graph.

b) Determine U_π and calculate the length L for $x = 10$ µm and $\lambda = 1.55$ µm.

c) Determine from the diagrams the values of T_{max} and T_{min} and deduce the extinction ratio of the Mach-Zehnder modulator in dB.

d) Calculate the ratio p in the Mach-Zehnder modulator.

HI1060

Performance Characterizations*

Nominal Delta (%)	1.06
Numerical Aperture	0.21
Refractive Index Value – Core	1.468 @ 850 nm
Core Diameter (µm)	3.4
Dispersion (ps/nm/km)	-65 @ 980 nm -50 @ 1060 nm

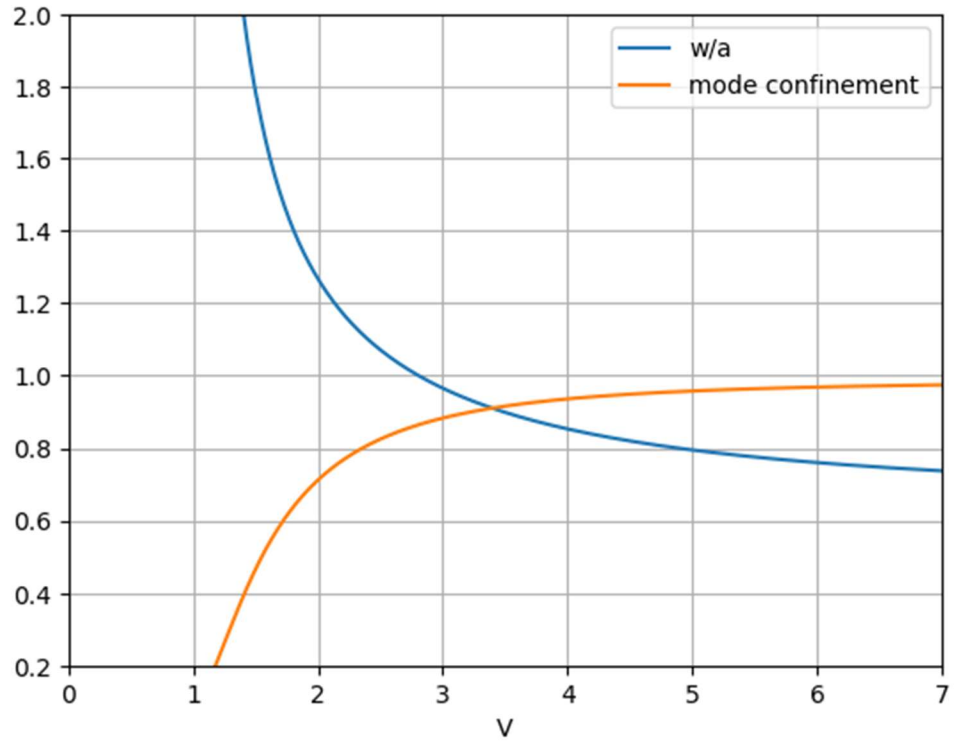
* Values in this table are nominal or calculated values

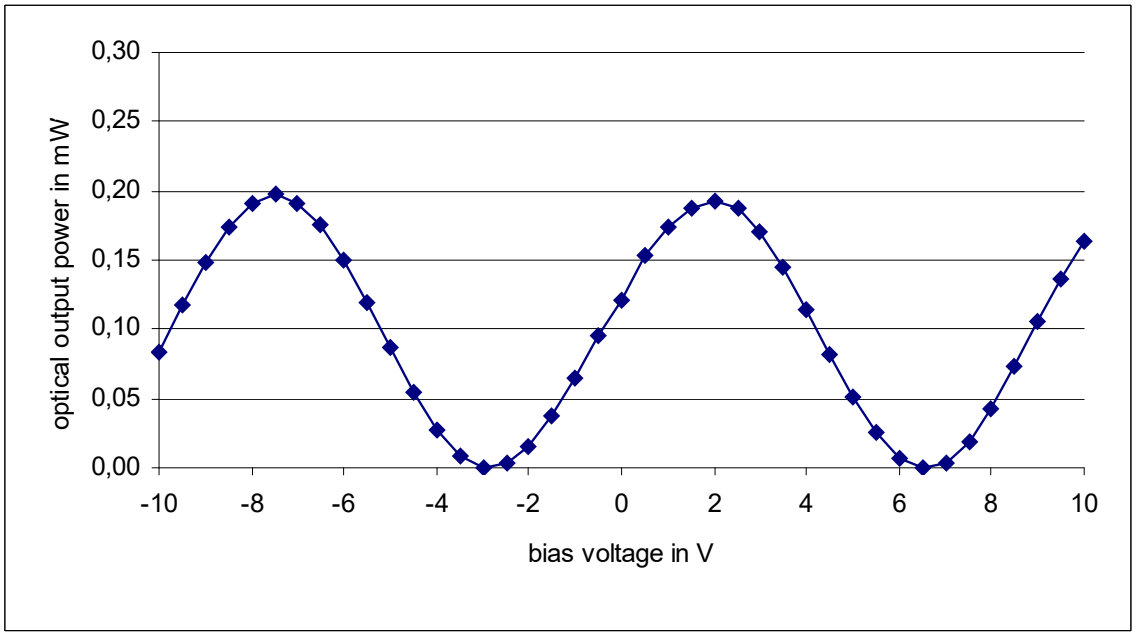
SMF28e+

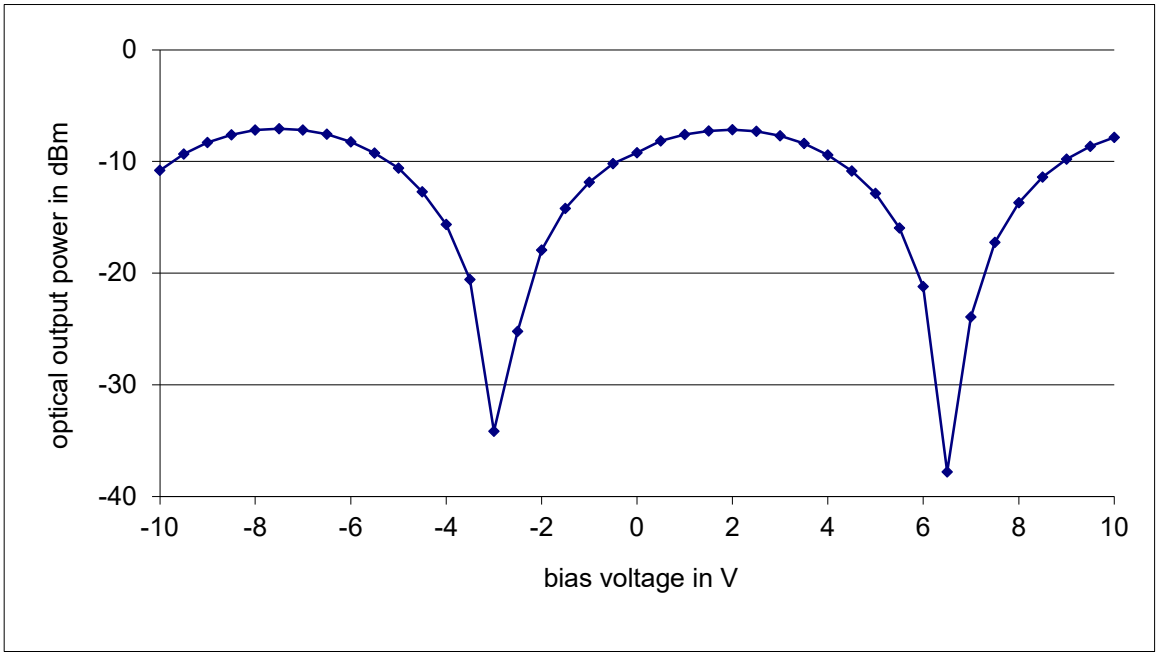
Leistungscharakterisierungen

Die charakterisierten Parameter sind typische Werte.

Kern-Durchmesser	8.2 µm
Numerische Apertur	0.14 NA wird bei einem einprozentigen Leistungspegel eines eindimensionalen Fernfeld-Scans bei 1310 nm gemessen.
Gruppenbrechzahl (n_{eff})	1310 nm: 1.4674 1550 nm: 1.4679
Ermüdungsbeständigkeit Parameter (n_d)	20
Absetzkraft der Beschichtung	Trocken: 0.6 lbs. (3 N) Nass, 14 Tage Raumtemperatur: 0.6 lbs. (3 N)
Rayleigh-Rückstreuungskoefffizient (für 1 ns Impulsbreite)	1310 nm: -77 dB 1550 nm: -82 dB




0,30
0,25
0,20
0,15
0,10
0,05
0,00
optical output power in mW
-10
-8
-6
-4
-2
0
2
4
6
8
10
bias voltage in V


0
-10
-20
-30
-40
optical output power in dBm
-10
-8
-6
-4
-2
0
2
4
6
8
10
bias voltage in V